D 32731 **(Pages : 2)** **Name.....**

Reg. No.....

FIRST SEMESTER M.Sc. DEGREE (REGULAR/SUPPLEMENTARY) EXAMINATION, NOVEMBER 2022

(CBCSS)

Physics

PHY 1C 02—MATHEMATICAL PHYSICS—I

(2019 Admission onwards)

Time : Three Hours Maximum : 30 Weightage

Section A

8 *Short questions answerable within* 7.5 *minutes)*
Answer **all** *questions, each question carries weightage* 1.

1. If V represents a vector derive the divergence of V in orthogonal curvilinear co-ordinates.
2. Is the given matrix Hermitian ?
3. Explain quotient law for tensors.
4. Explain the general form of a second order differential equation and classify them based on being elliptic, parabolic or hyperbolic.
5. Explain the significance of the Dirac-Delta function.
6. Explain Gram-Schmidt orthogonalization.
7. Evaluate the Fourier co-efficients a_0, a_n and b_n for a piecewise continuous function where L is half of the period of the function.
8. Explain the convolution property of Fourier transform with an example.

(8 × 1 = 8 weightage)

Section B

4 *essay questions answerable within* 30 *minutes*
Answer any **two** *questions, each question carries weightage* 5

9. Explain the Frobenius' method of finding solution to homogenous differential equation of second order.
10. Using appropriate differential equation obtain the general form for Hermite polynomials of degree n.

Turn over

11. Define a Fourier transform. Explain any *five* properties of Fourier transforms.

12. Explain the algebraic operations of Tensors.

(2 × 5 = 10 weightage)

Section C

7 problems answerable within 15 minutes

Answer any **four** *questions, each question carries weightage* 3.

13. A string of length π is stretched until the wave speed is 40 m/sec. It is given an initial velocity of $4 \sin(x)$ from its initial position. What is maximum displacement ?

14. Evaluate $\Gamma\left(-\frac{3}{2}\right)$.

15. If A_{ij} is antisymmetric tensor, find the component A_{11}.

16. Find Laplace transform of the function $F(t) = \frac{e^{at} - 1}{a}$.

17. For the Legendre polynomial prove that $P_n(x) = 1$.

18. Find the Fourier series of the function e^x in the interval $-\pi < x < \pi$.

19. Prove that the given matrix is unitary :

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}.$$

(4 × 3 = 12 weightage)